

Summary Statistics of UAD PUF

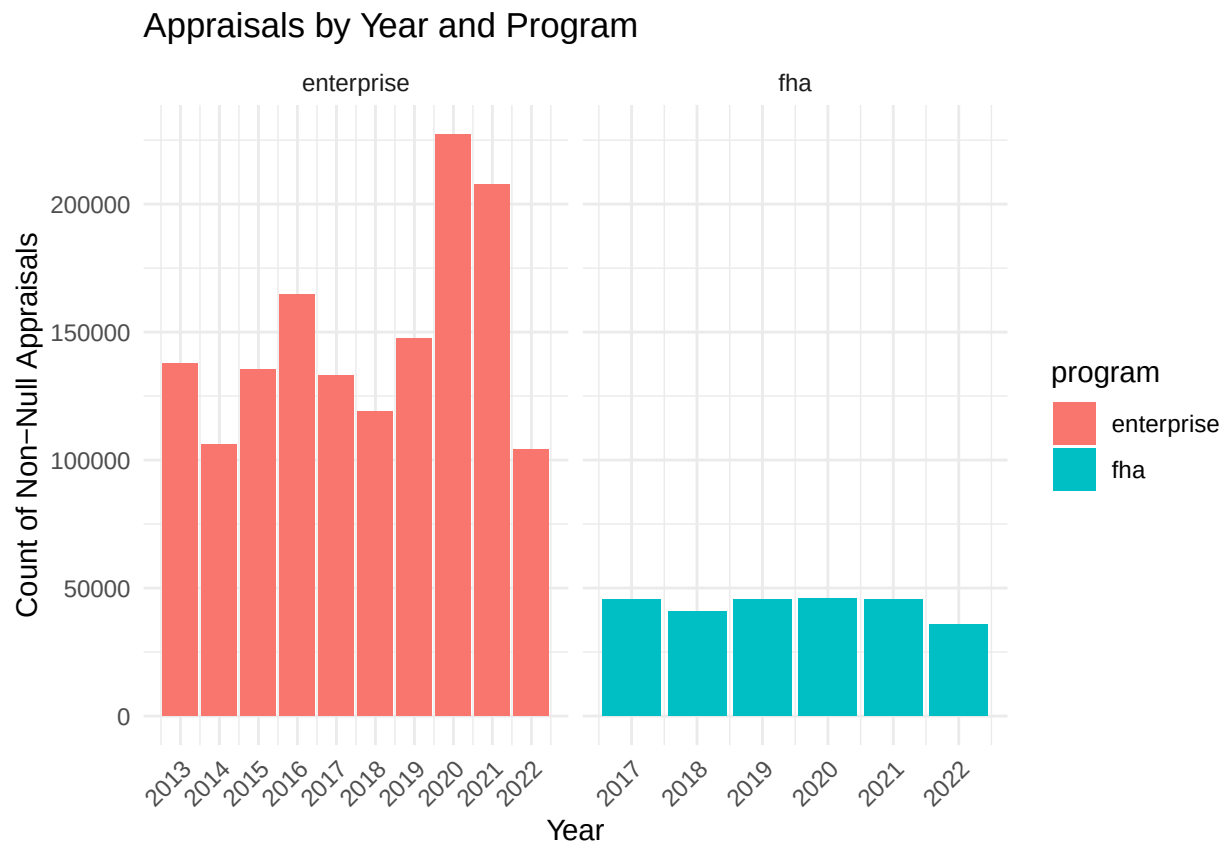
2026-09-24

Observations over time

```
# Load dataset

df <- read_dta(here("data", "clean", "uad_clean.dta"))

df %>%
  group_by(program, year) %>%
  summarise(non_null_appraisals = sum(!is.na(appraised_value)), .groups = "drop") %>%
  ggplot(aes(x = year, y = non_null_appraisals, fill = program)) +
  geom_col() +
  scale_x_continuous(breaks = 2013:2022) +
  facet_wrap(~ program, scales = "free_x") +
  labs(x = "Year", y = "Count of Non-Null Appraisals", title = "Appraisals by Year and Program") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Geographic summary stats

```
df %>%
  summarise(across(c("state_fips", "county_fips", "tract_fips"), ~ n_distinct(.x, na.rm = TRUE))) %>%
  knitr::kable()
```

state_fips	county_fips	tract_fips
52	3134	77383

```
df %>%
  count(state_fips, sort = TRUE) %>%
  head(10) %>%
  mutate(state_name = fips_info(state_fips)$full) %>%
  select(state_fips, state_name, count = n) %>%
  knitr::kable()
```

state_fips	state_name	count
06	California	209512
48	Texas	164167
12	Florida	123042
13	Georgia	61682
37	North Carolina	60622
17	Illinois	59859
04	Arizona	57956
42	Pennsylvania	55913
39	Ohio	54186
08	Colorado	53776

Appraisal summary stats

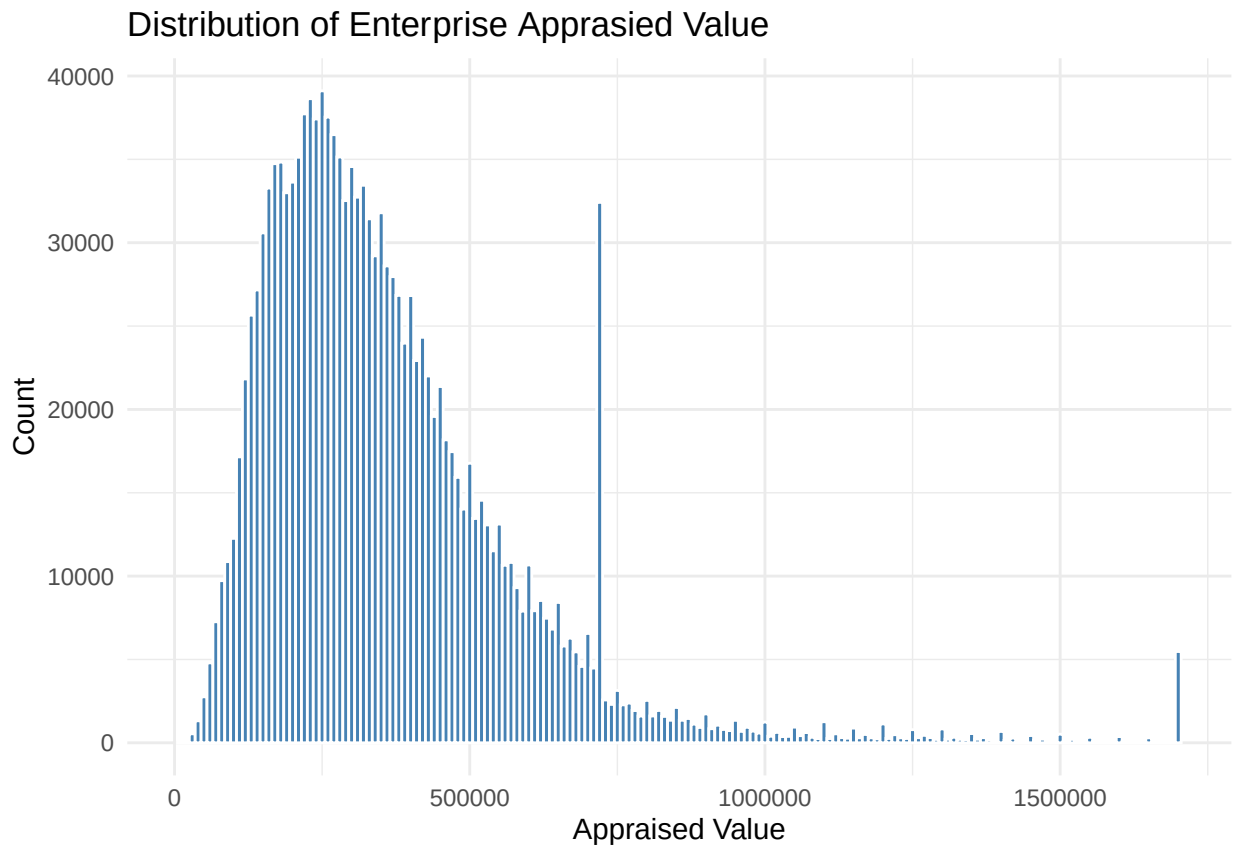
```
df %>%
  select(program, appraised_value, contract_price) %>%
  pivot_longer(cols = c(appraised_value, contract_price), names_to = "Variable", values_to = "Value") %>%
  filter(!is.na(Value)) %>%
  group_by(Variable, program) %>%
  summarise(
    N = n(),
    Mean = mean(Value),
    `10th Percentile` = quantile(Value, 0.10),
    Median = median(Value),
    `90th Percentile` = quantile(Value, 0.90),
    .groups = "drop"
  ) %>%
  knitr::kable(format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Variable	program	N	Mean	10th Percentile	Median	90th Percentile
appraised_value	enterprise	1,481,868	369,936.4	155,000	325,000	645,000
appraised_value	fha	259,265	254,918.5	125,000	235,000	415,000
contract_price	enterprise	746,237	336,383.1	145,000	305,000	575,000

contract_price	fha	209,649	243,067.6	115,000	225,000	385,000
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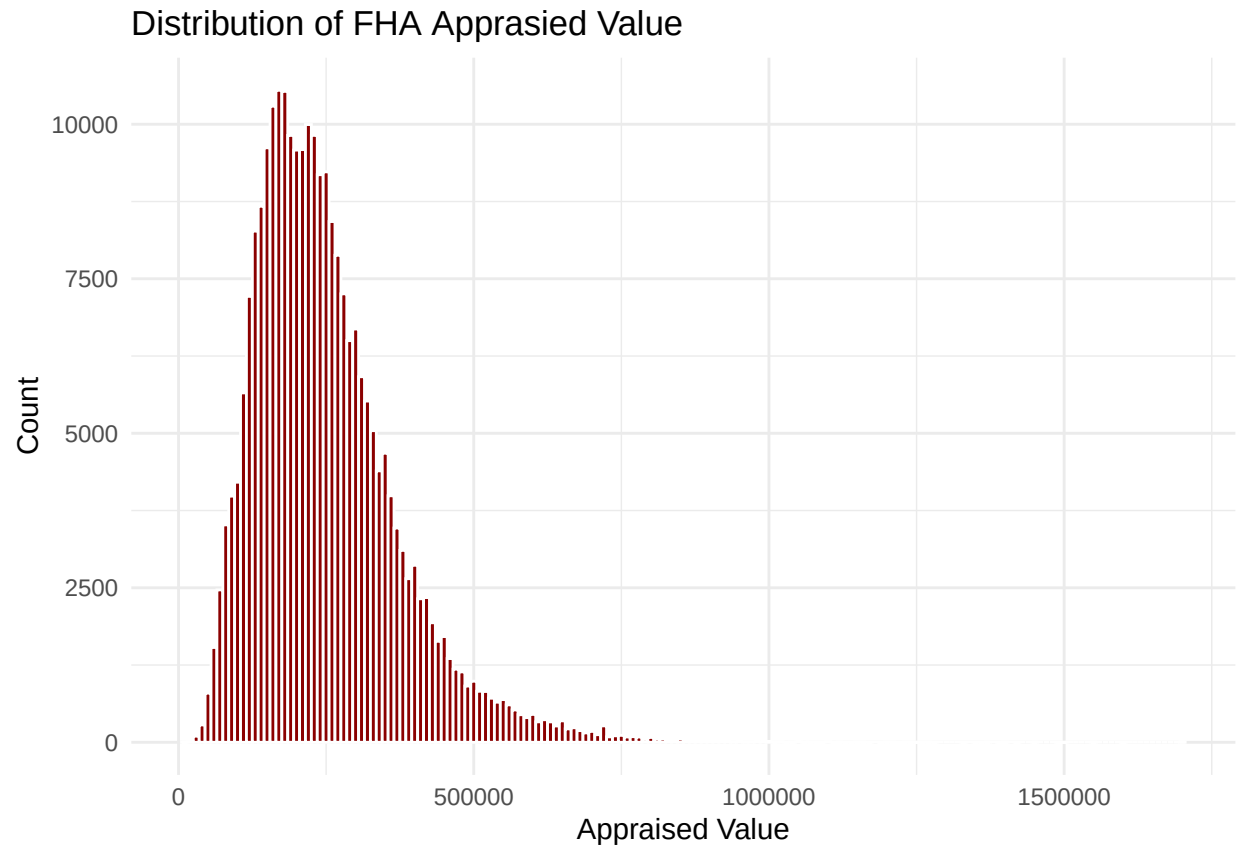
```
df %>%
  filter(program == "enterprise", !is.na(appraised_value)) %>%

  ggplot(aes(x = appraised_value)) +
    geom_histogram(binwidth = 10000, fill = "steelblue", color = "white") +
    labs(
      x = "Appraised Value",
      y = "Count",
      title = "Distribution of Enterprise Apprasied Value") +
    theme_minimal()
```



```
df %>%
  filter(program == "fha", !is.na(appraised_value)) %>%

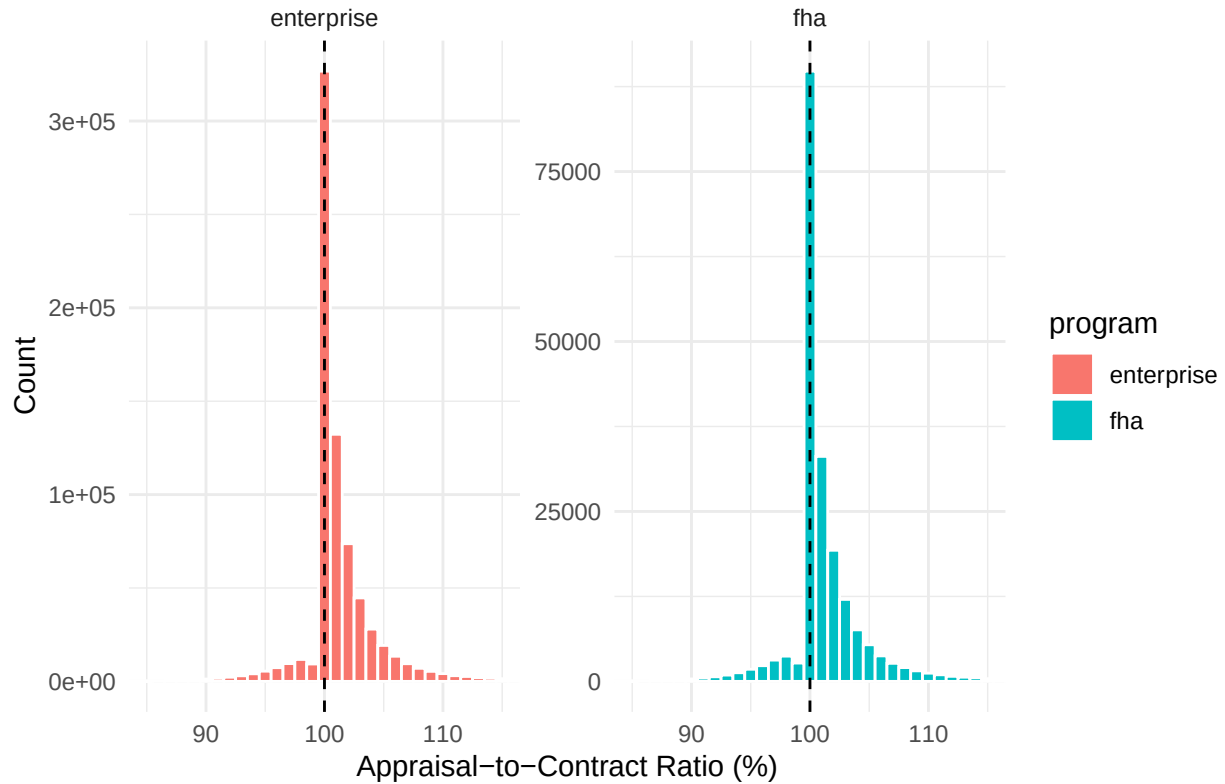
  ggplot(aes(x = appraised_value)) +
    geom_histogram(binwidth = 10000, fill = "darkred", color = "white") +
    labs(
      x = "Appraised Value",
      y = "Count",
      title = "Distribution of FHA Apprasied Value") +
    theme_minimal()
```



Appraisal to contract ratio

```
df %>%
  filter(!is.na(appraisal_to_contract)) %>%
  ggplot(aes(x = appraisal_to_contract, fill = program)) +
    geom_histogram(binwidth = 1, color = "white") +
    geom_vline(xintercept = 100, linetype = "dashed", color = "black") +
    facet_wrap(~ program, scales = "free_y") +
    coord_cartesian(xlim = c(85, 115)) +
    labs(x = "Appraisal-to-Contract Ratio (%)", y = "Count",
         title = "Appraised Value as a Percentage of Contract Price") +
    theme_minimal()
```

Appraised Value as a Percentage of Contract Price



Property characteristics

```
df %>%
  select(quality, condition, bedrooms, bathrooms, lot_size, gross_living_area) %>%
  mutate(across(everything(), as_factor)) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(x = Value)) +
    geom_bar(fill = "steelblue") +
    facet_wrap(~ Variable, scales = "free") +
    scale_y_continuous(labels = scales::comma) +
    theme_minimal() +
    labs(title = "Distribution of Property Characteristics")
```

Distribution of Property Characteristics

